

Reactive Oxygen Species, Aging, and the Science of Balance

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युक्तं युक्तं त्यजेद् वायुं युक्तं युक्तं च पूरयेत् । युक्तं युक्तं च बध्नीयाद् एवं सिद्धिमवाप्नुयात् ॥

--- हठयोगप्रदीपिका - २.१८

One should exhale properly, inhale properly, and retain the breath properly; thus mastery is attained.

Every cell in the human body survives by using oxygen. Oxygen allows us to extract energy from food, power cellular functions, and sustain life itself. Yet this same oxygen, when handled inefficiently, becomes a source of damage. This paradox lies at the heart of modern theories of aging, inflammation, and chronic disease, and it revolves around a group of molecules known as reactive oxygen species (ROS). These are chemically reactive byproducts generated during normal cellular metabolism, especially within the mitochondria. In controlled amounts, ROS are essential for cell signalling, immune defence, and physiological adaptation. The problem begins when their production exceeds the body's capacity to neutralize them. When ROS production exceeds the body's ability to neutralize them, a state known as oxidative stress develops. In this state, reactive molecules begin damaging cell membranes, proteins, enzymes, and even DNA. Over time, this damage accumulates silently, contributing to accelerated aging, chronic inflammation, metabolic disorders, neurodegeneration, and reduced immunity.

What is often overlooked, however, is why ROS production becomes excessive in the first place. It is not merely a consequence of aging, pollution, or poor diet. A major driver is inefficiency - inefficient metabolism, inefficient oxygen utilization, and inefficient regulation of stress. This is where breathing becomes central.

When the body is under constant mental or emotional stress, breathing patterns change. Breath becomes shallow, rapid, and irregular. Oxygen may enter the lungs, but its delivery to tissues becomes poorly regulated. Mitochondria are forced to work under strained conditions, producing energy with greater leakage in the form of reactive oxygen species. In simple terms, a stressed body burns fuel poorly and produces more internal waste. Deep, slow breathing reverses this inefficiency. By slowing respiratory rate and increasing diaphragmatic involvement, deep breathing improves gas exchange and stabilizes carbon dioxide levels, which in turn enhances oxygen release to tissues. When cells receive oxygen steadily rather than erratically, mitochondrial function becomes more efficient and ROS generation naturally decreases. Thus, deep breathing does not merely increase oxygen - it prevents oxygen from becoming harmful.

Āyurvedic lens to the breathing

In Ayurvedic thought, excessive internal damage arises when Vāta Doṣa becomes aggravated - when movement becomes excessive, irregular, and uncontrolled. Vāta governs respiration, nervous activity, circulation, and cellular communication. When vāta is disturbed, internal friction increases. Heat, dryness, instability, and degeneration follow. This description aligns closely with what modern science identifies as oxidative stress. A calm mind does not constantly push the body into anticipatory stress, and a settled nervous system allows repair processes to proceed without interruption.

Ayurveda also emphasized external strategies to neutralize internal stress. One such practice is earthing, or direct contact with the earth. Modern research shows that the Earth's surface carries a continuous supply of free electrons, and that direct contact allows the human body - an electrical system - to absorb these electrons. These electrons can neutralize reactive oxygen species, acting as natural antioxidants. Ayurveda did not use electrical terminology, yet it consistently advised contact with soil, stone, water, and the ground to stabilize vāta and reduce internal agitation. What science now measures electronically, Ayurveda observed experientially.

Food plays an equally crucial role. Modern nutrition speaks of antioxidants - molecules that neutralize reactive oxygen species. Ayurveda described the same principle through rasa, vīrya, and vipāka, recommending fresh, seasonal, naturally colorful foods (Fruits, vegetables, herbs, spices) that support tissue nourishment and prevent degeneration.

Importantly, Ayurveda sees deep breathing, grounding, mindful eating, exposure to sunlight, rhythmic daily routines, and adequate rest as parts of a single system of self-regulation. When these are aligned, oxidative stress

does not need to be aggressively fought - it remains naturally contained. Deep breathing, therefore, emerges as a central practice - not because it is simple, but because it is fundamental. It slows the mind, stabilizes metabolism, reduces unnecessary oxidative reactions, and preserves cellular integrity. In doing so, it slows biological aging by reducing internal wear.

Breath, Sound, and Indian Culture

Indian culture trained mindful breathing through lived practices. The deep inhalation of flowers and natural fragrances was never merely aesthetic - it uplifted prāṇa and refined sensory perception. Activities such as Chanting, singing, mantra, Vedic recitation, humming, and ritual sound repeatedly coordinated breath, voice, rhythm, and attention. The Chāndogya Upaniṣad makes this relationship explicit:

अथ खलु व्यानमेवोद्गीथमुपासीत यद् वै प्राणिति स प्राणो यदपानिति सोऽपानः । अथ यः प्राणापानयोः संधिः स व्यानो यो व्यानः सा वाक् । तस्मादप्राणन्नपानन् वाचमभिव्याहरति ॥

--- छान्दोग्योपनिषत् - १.३.३

What is breathed is prāṇa, what is breathed out is apāna; their junction is vyāna, and vyāna is speech.

The verse above thus treats speech as inseparable from the regulation of breath. This provides an important key to understanding Indian traditions of chanting and recitation. The architecture of Sanskrit recitation encourages a disciplined coordination of breath and sound. When chanting is rhythmic and sustained, the reciter naturally develops regular cycles of inhalation and exhalation, controlled phonation, pauses, and prolonged expiration. Studies of rhythmic mantra recitation have shown that such practices can slow and regularize breathing and synchronize respiratory and cardiovascular rhythms. Yoga texts even speak of humming, using both breath and sound, such as :

अनिलं मन्दवेगेन भ्रामरीकुम्भकं चरेत् । मन्दं मन्दं रेचयेद् वायुं भृङ्गनादं ततो भवेत् ॥

--- घेरण्डसंहिता - ५.७८

Practise Bhāmārī with a gentle breath; slowly, slowly release the breath, producing the sound of a bee.

Finally, deep breathing also improves the mechanics of pulmonary gas exchange. When the breath is drawn fully with the diaphragm and abdomen, a larger tidal volume reaches the alveoli, increasing the proportion of fresh air available for exchange. Oxygen can therefore be taken up efficiently across the alveolar-capillary membrane, while carbon dioxide carried in the blood from metabolically active tissues is more effectively eliminated during a complete exhalation. In shallow breathing, a greater proportion of each breath is confined to the anatomical dead space, so alveolar ventilation can become less efficient. The result is less effective oxygen uptake and CO₂ elimination for a given minute ventilation.

प्राणं सूर्येण चाकृष्य पूरयेदुदरं शनैः । विधिवत्कुम्भकं कृत्वा पुनश्चन्द्रेण रेचयेत् ॥

--- हठयोगप्रदीपिका - २.८

Drawing in the prāṇa, one should slowly fill the abdomen; retaining it properly, one should exhale again.

This matters at the cellular level. Mitochondria require oxygen for efficient oxidative phosphorylation, but electron transport also inevitably produces reactive oxygen species. When oxygen delivery, CO₂ regulation, autonomic activity, and cellular metabolism are poorly coordinated, mitochondrial electron leakage and oxidative stress can increase. Deep, slow and well-regulated breathing helps optimize respiratory gas exchange and autonomic-metabolic regulation, thereby reducing excessive mitochondrial ROS generation.

The crucial word is शनैः - slowly, gradually. Classical Yoga therefore did not speak of “deep breathing” in modern physiological language, but it clearly prescribed deliberate, controlled and gradual regulation of the breath. This is a remarkably compact description of what modern physiology would recognize as controlled respiration with prolonged vocalized exhalation. The conch, chanting, humming, and singing can similarly be understood as traditional activities that repeatedly engaged respiratory muscles and regulated the breath. Modern respiratory-muscle training likewise uses resistance, deep inhalation, and controlled exhalation to strengthen respiratory function.

Deep breathing therefore emerges as one of the simplest tools of this larger philosophy since it requires no equipment and can be integrated into ordinary life.